

Qns	Answer	Marks
2(a)(i)	$s = ut + \frac{1}{2}at^2$ (Taking upwards as positive) $-7.8 = (5.9 \sin 60^\circ)t + \frac{1}{2}(-9.81)t^2$ $t = 1.9 \text{ s (or 1.89 s)}$	C1 A1
2(a)(ii)	Vertically: (Taking upwards as positive) $v^2 = u^2 + 2as$ $v_v^2 = (5.9 \sin 60^\circ)^2 + 2(-9.81)(1.2 - 9.0)$ $v_v = -13.4 \text{ m s}^{-1}$ or $v = u + at$ $= -5.9 \sin 60^\circ + 9.81(1.885)$ $= -13.4 \text{ m s}^{-1}$ (Correct substitution and answer) horizontally: $v_h = 5.9 \cos 60^\circ$ $v_h = 2.95$ (Correct substitution and answer) $\tan \theta = \frac{13.4}{2.95}$ $\theta = 78^\circ$	C1 C1 A1
2(b)(i)	speed decrease , (So) viscous force / drag (force) decrease (resultant force decreases as) upthrust and weight remain the same	B1 B1
2(b)(ii)	$U = \rho Vg$ $U = (1000)(9.81)(7.5 \times 10^{-2})$ $= 736 \text{ N}$ $= 740 \text{ N}$	M1
2(b)(iii)	resultant force = 740 + 950 – (78)(9.81) = 925 acceleration = F / m = 925 / 78 = 12 m s ⁻² (vertically) upwards	C1 A1 A1

